# NeST SE(3) decode tree on one real YCB-V image: the beam-MAP path through the joint tree

each level is a 64-way child = 8 rotations (Mollweide cards lifted into 3D; HEALPix cells = $S^2$ binning, hue = Hopf roll) × 8 translation octants (octree cube)

**gold = the beam-MAP root → leaf path (branch width ∝ joint probability): the cell refines rotation ×8 and translation ×8 each level**

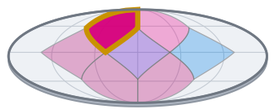



actual input
(obj 3: sugar box)

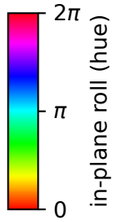
**ViT-B/16 encoder**

*tokens condition every node*

**root** $p_0(c_0 \mid x)$
**72-way rotation**








**trans octant**
(+ − −) $p = 0.90$

$\ell = 1$

64-way
8 × 8
nside 2

rot $p = 0.00$ | rot $p = 0.00$ | **rot** $p = 0.64$ | rot $p = 0.00$ | rot $p = 0.00$ | rot $p = 0.00$ | rot $p = 0.34$ | rot $p = 0.02$

**trans octant**
(− + +) $p = 1.00$

$\ell = 2$

64-way
8 × 8
nside 4

rot $p = 0.05$ | rot $p = 0.00$ | rot $p = 0.03$ | rot $p = 0.00$ | rot $p = 0.59$ | rot $p = 0.04$ | **rot** $p = 0.28$ | rot $p = 0.01$

**trans octant**
(− + +) $p = 0.98$

$\ell = 3$

64-way
8 × 8
nside 8

rot $p = 0.20$ | rot $p = 0.19$ | rot $p = 0.01$ | rot $p = 0.00$ | rot $p = 0.20$ | **rot** $p = 0.36$ | rot $p = 0.03$ | rot $p = 0.00$



ground truth — NeST MAP

**leaf node =**
**output 6D pose → 3D box**
**scene 51/1629: rot** 6.8°, **trans 8 mm**